



Hand in electronically in the form of a PDF-file before 20.00. You may freely use formulas from the formula sheet on the canvas page. Refer to relevant theorems from the course when you use them. Write clearly with short and concise motivations.

1. In each case determine whether or not the series converges

$$\text{a) } \sum_{n=1}^{\infty} (-1)^n \cos \frac{1}{n} \quad , \quad \text{b) } \sum_{n=1}^{\infty} \frac{\sin(\frac{n\pi}{2})}{\sqrt{n}} \quad , \quad \text{c) } \sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}.$$

2. Let $f(x)$ be odd, 2π -periodic, and satisfy $f(x) = \cos x$ when $0 < x < \pi$. Compute the Fourier series of f and use the result to compute the sum

$$\sum_{n=1}^{\infty} \frac{n^2}{(4n^2 - 1)^2}.$$

3. Find a solution $u(x, t)$ for $0 \leq x \leq \pi/2$, $t \geq 0$ to the following problem for the heat equation:

$$\begin{cases} u_t = 3u_{xx}, & 0 < x < \pi/2, t > 0, \\ u(0, t) = u_x(\pi/2, t) = 0, & t > 0, \\ u(x, 0) = \cos 2x \sin x, & 0 < x < \pi/2. \end{cases}$$

4. Seek a power series solution to the problem

$$xy''(x) + 2y'(x) + xy(x) = 1, \quad y(0) = 0.$$

5. Consider the function $g(x) = e^{-|x|} \operatorname{sign} x$ where the sign-function $\operatorname{sign} x$ is $+1$ if $x > 0$ and -1 if $x < 0$.

a) Find the Fourier transform of g .

b) Determine whether or not there exists a function $f \in L^1(\mathbb{R})$ such that for all $x \in \mathbb{R}$

$$\int_{-\infty}^{+\infty} f(x-t)g(t) dt = xe^{-x^2/2}.$$

Also, if a solution f exists, try to determine one.

6. Consider the functions $f_n(x) = ne^{-nx} + \sin nx$ for $n = 1, 2, \dots$, and let $g(x)$ be a continuous function in $L^1(\mathbb{R})$. Find the limit

$$\lim_{n \rightarrow \infty} \int_0^{\infty} g(x)f_n(x) dx.$$